Kenny Chen

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Education

McMaster University Hamilton, ON
Bachelor of Applied Science (BASc) in Computer Engineering *Expected 2029*

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, C, C++, SQL, Swift
Technologies: React, React Native, Node.js, Git, CMake, MATLAB, PyTorch, OpenGL, ESP32, SwiftUI, CoreBluetooth, Selenium, Catch2, FastAPI, Autodesk Inventor, SolidWorks

Projects

MiniGrad: Deep Learning Framework | *C++* Aug. 2026 – Present
- Building a C++ deep learning framework modeled on PyTorch's c10 core; storage layer implemented, tensor layer and autograd in progress.
- Implemented the storage layer: device abstraction, type-erased data pointers with custom deleters, a CPU allocator, and intrusive reference counting for tensor storage.

ROSIE: Ball Launcher and iOS App for Individuals with Arthritis | *C++, Swift, Inventor* Apr. 2026
- Built the iOS app and ESP32 BLE pipeline on a 5-person team designing an under-$200 tennis ball launcher for a client with rheumatoid arthritis.
- Implemented hands-free voice control with SFSpeechRecognizer, sending recognized commands to the ESP32 through CoreBluetooth over a custom GATT service.
- Wired the power and drive system: battery to a buck converter supplying the ESP32, with motor driver modules running a 775 DC motor.
- Modeled the main frame, buck converter cover, and handles in Autodesk Inventor.

Minecraft Clone: Voxel Game Engine | *C++, OpenGL, CMake* May 2025 – Present
- Building a multithreaded voxel engine with OpenGL. Chunk generation and meshing are handled on worker threads, while GPU uploads are on the main thread.
- Bit-packed vertex attributes from 24 bytes to 4 bytes, generating frames 1.4× faster compared to unpacked vertices due to reduced GPU bandwidth overhead.
- Applied additional CPU and rendering optimizations including prioritizing cache hits, culling hidden faces, and instancing draw calls.

Best AI Hack, YRHacks 2025: Google Reviews Scraper | *React, FastAPI, Selenium* Apr. 2025
- Worked in a 3-person team to develop a program that takes the URL of a restaurant on Google, scrapes the reviews, and aggregates these results through an LLM for a concise summary. Won against 1200+ competitors.
- Built the Selenium scraping layer, FastAPI backend, React frontend, and summarization pipeline.

Experience

Busboy Oct. 2024 – Aug. 2025
Longing Fusion Cuisine *Markham, ON*
- Managed up to 6 tables per shift simultaneously, prioritizing tasks to ensure timely and accurate order delivery.
- Accurately handled cash, credit, and split-bill transactions, maintaining a balanced till at end of each shift.

Awards

DECA 2025 ICDC Qualifier | *Buying and Merchandising Operations Research* Oct. 2024 – Feb. 2025
- Qualified for ICDC with a 2-person team out of 6900+ Ontario competitors, proposing AI automation strategies for Harvey's drive-thru, staffing, and social media.
- Wrote a 20-page business report and pitched it to judges.